

Instagram Data Analysis Using Python

Internship Task 3: Alfido Tech

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Abstract

This project focuses on analyzing Instagram data to understand user engagement, posting behavior, and content performance. The analysis was performed using multiple datasets containing information about users, photos, likes, comments, follows, and hashtags. Data cleaning and exploratory data analysis (EDA) techniques were applied to identify trends such as active users, popular content types, hashtag usage, and engagement patterns. The insights obtained from this analysis help improve social media marketing strategies and provide recommendations for increasing audience engagement on the Alfido Tech platform.

Objective

- To analyze Instagram user activity and engagement.
- To identify the most active users and popular content types.
- To study likes, comments, and hashtag usage.
- To understand posting patterns and user interactions.
- To provide business recommendations for improving Instagram engagement for Alfido Tech.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Kaggle Dataset
- Google Colab

Dataset

The dataset was collected from Kaggle and consists of seven CSV files representing different aspects of Instagram data.

- users.csv

- photos.csv
- likes.csv
- comments.csv
- follows.csv
- tags.csv
- photo_tags.csv

These datasets contain information about Instagram users, uploaded photos, likes, comments, followers, hashtags, and tagged photos, which were used for comprehensive analysis.

Data Cleaning

The dataset was cleaned before analysis to improve data quality and ensure accurate results. Duplicate records were removed, missing values were checked, and unnecessary data was handled appropriately. Date and time columns were converted into proper datetime format for easier analysis. The datasets were verified for consistency before performing exploratory data analysis and visualization.

Visualizations:

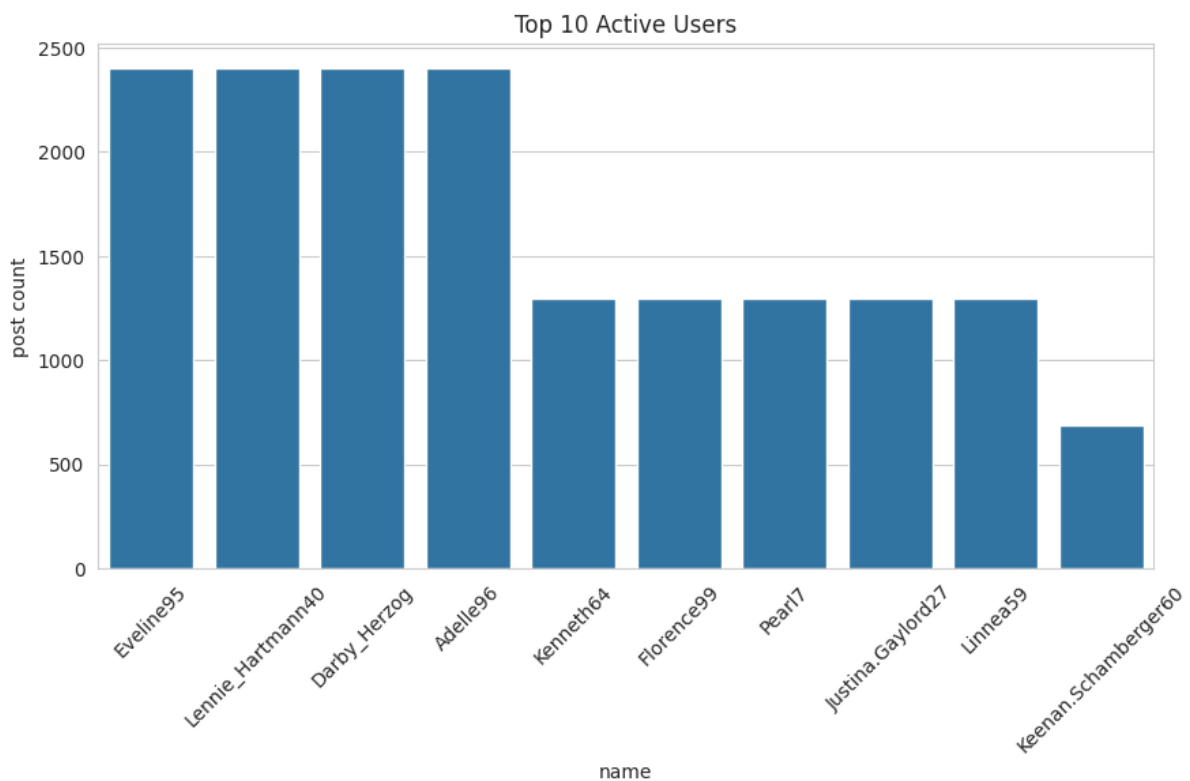


Fig 1.1:Top 10 Active Users

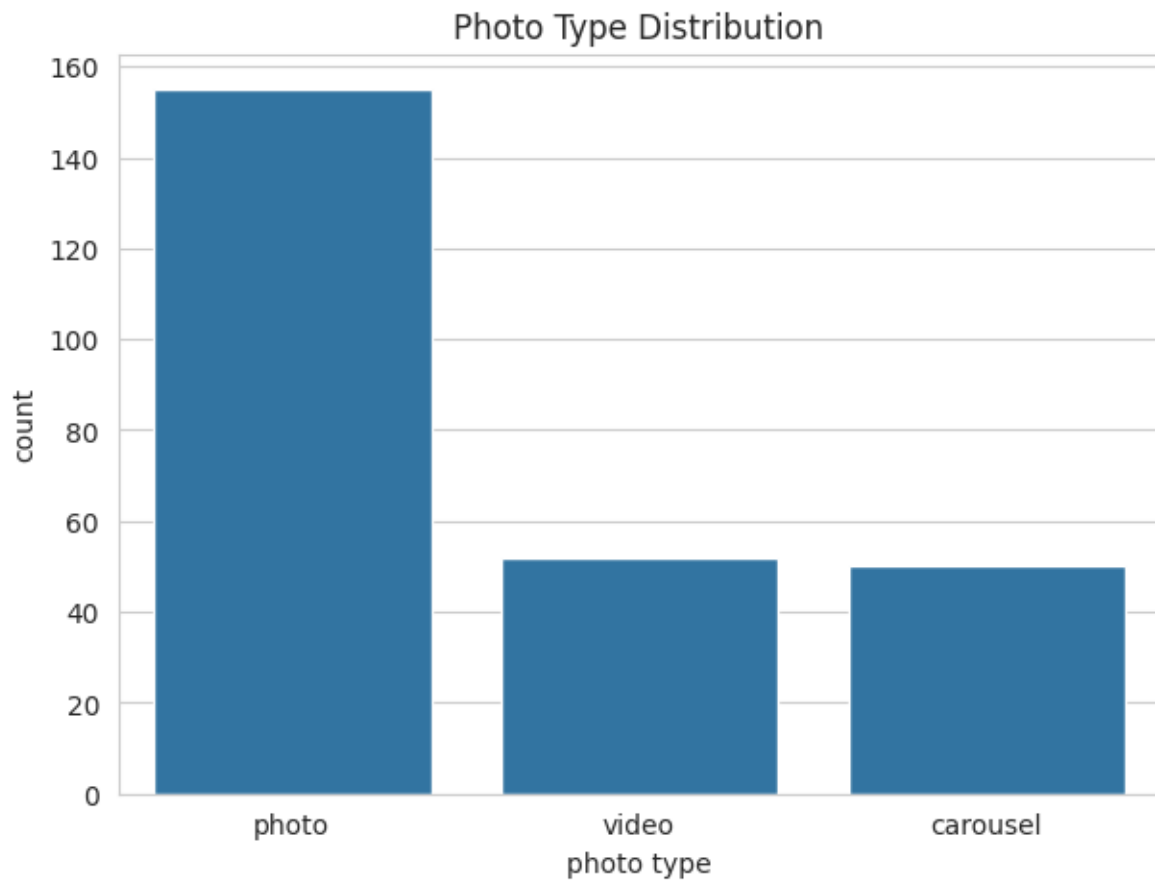


Fig 1.2:Photo Type Distribution

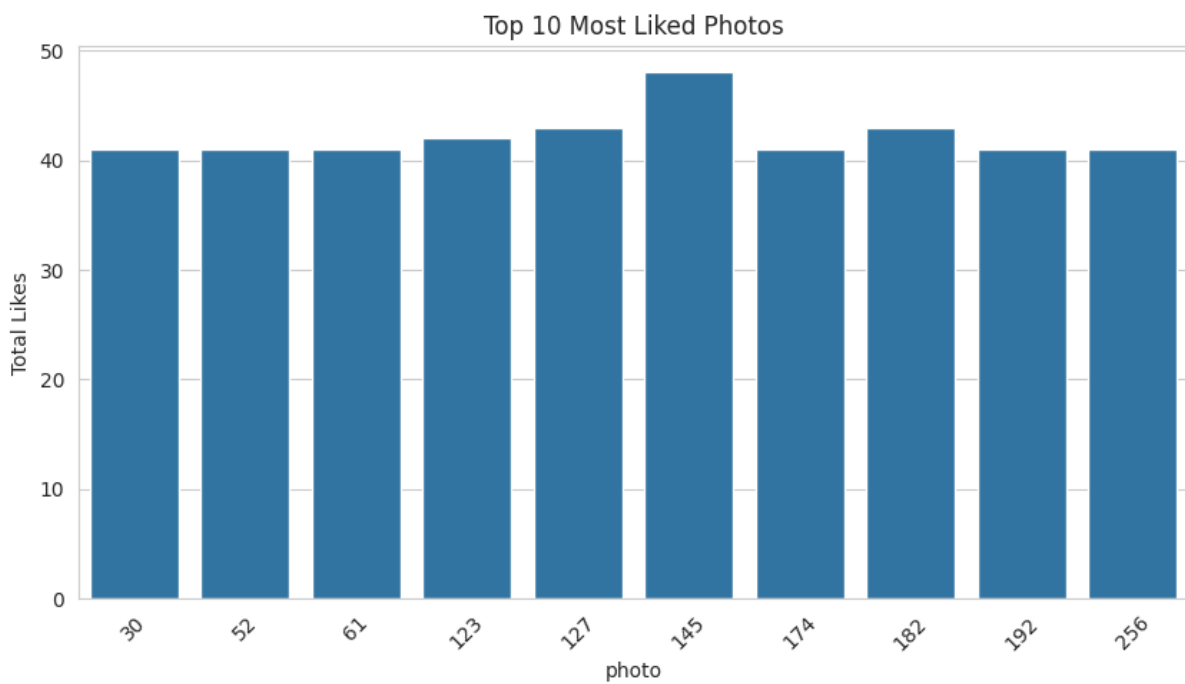


Fig 1.3:Top 10 Most Liked Photos

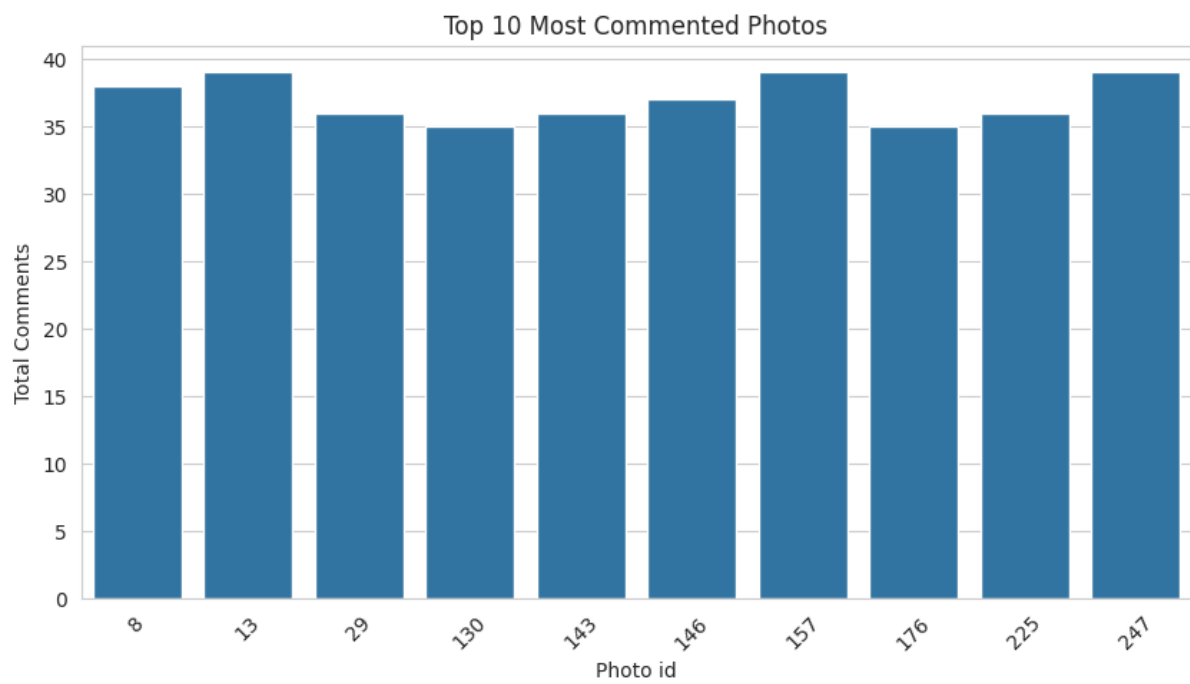


Fig1.4:Top 10 Most Commented Photos

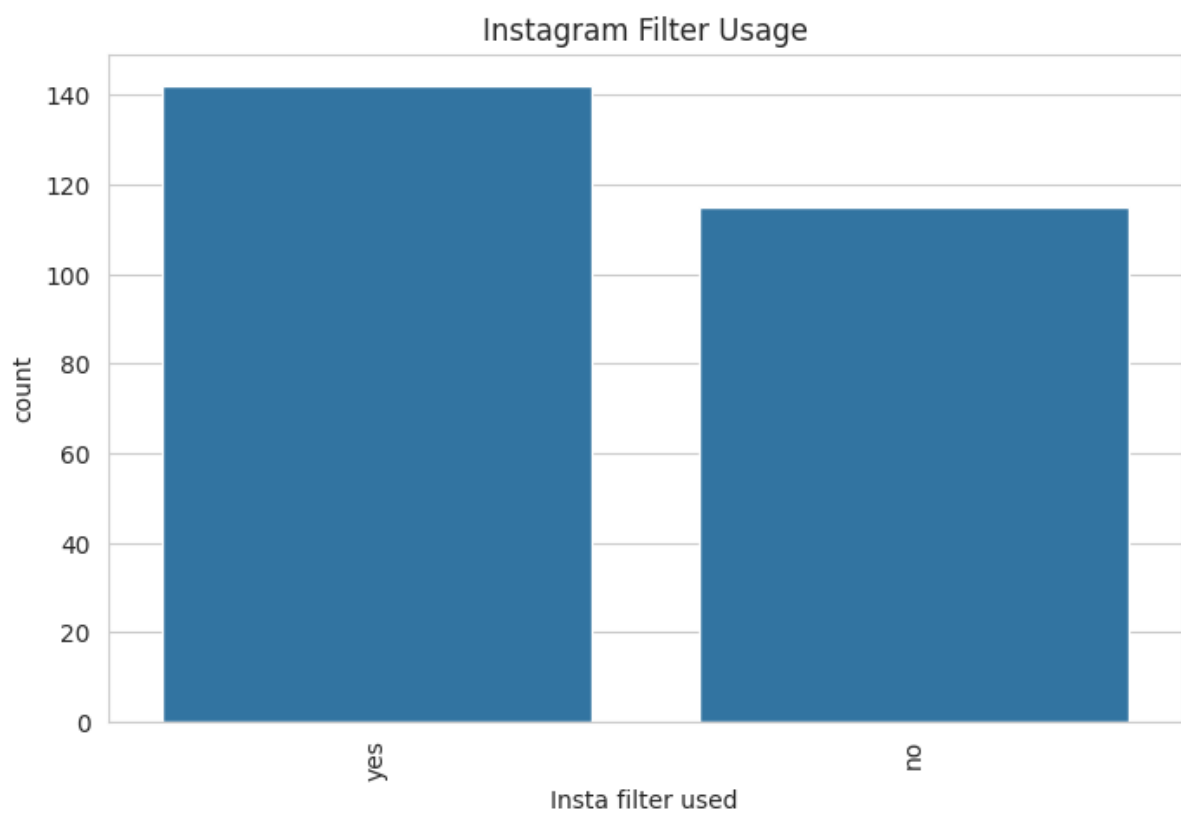


Fig1.5:Instagram Filter Usage

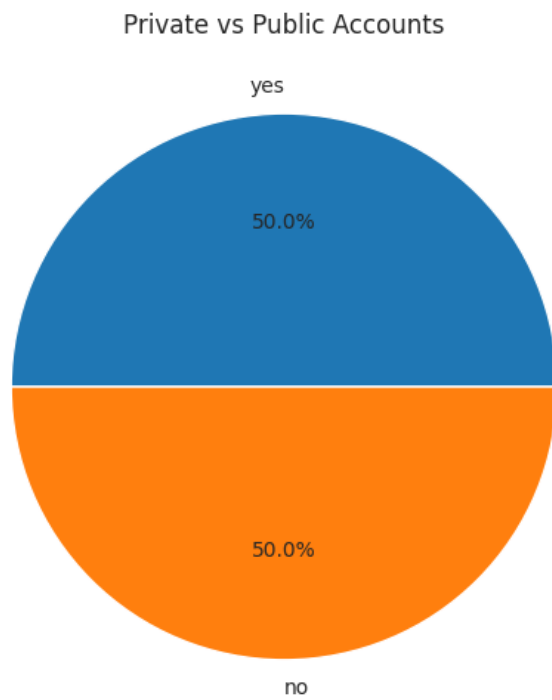


Fig 1.6: Private vs Public Accountnts

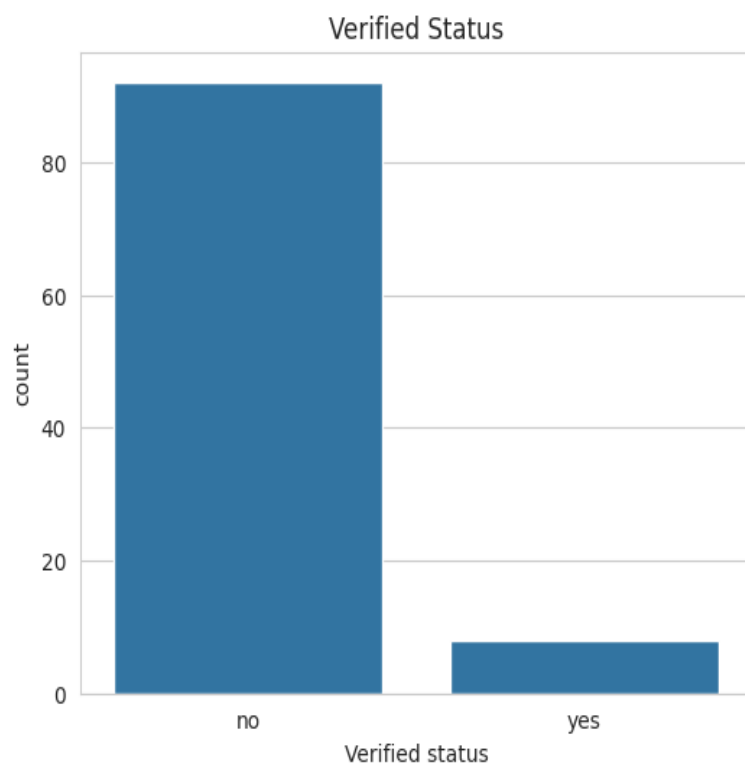


Fig 1.7: Verified Status

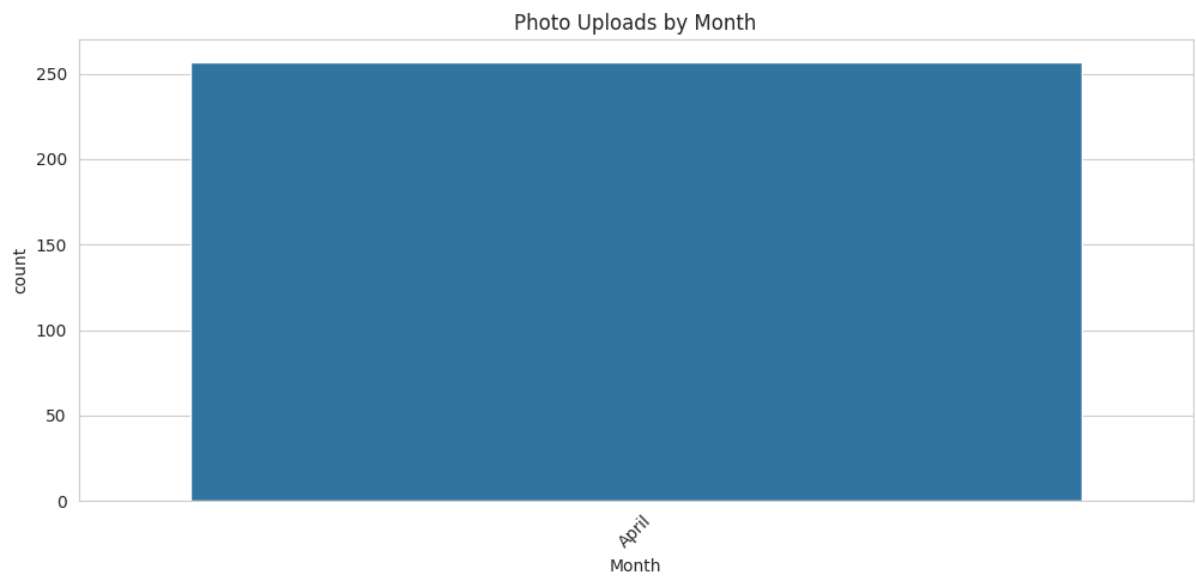


Fig 1.8:Photo Uploads by Month

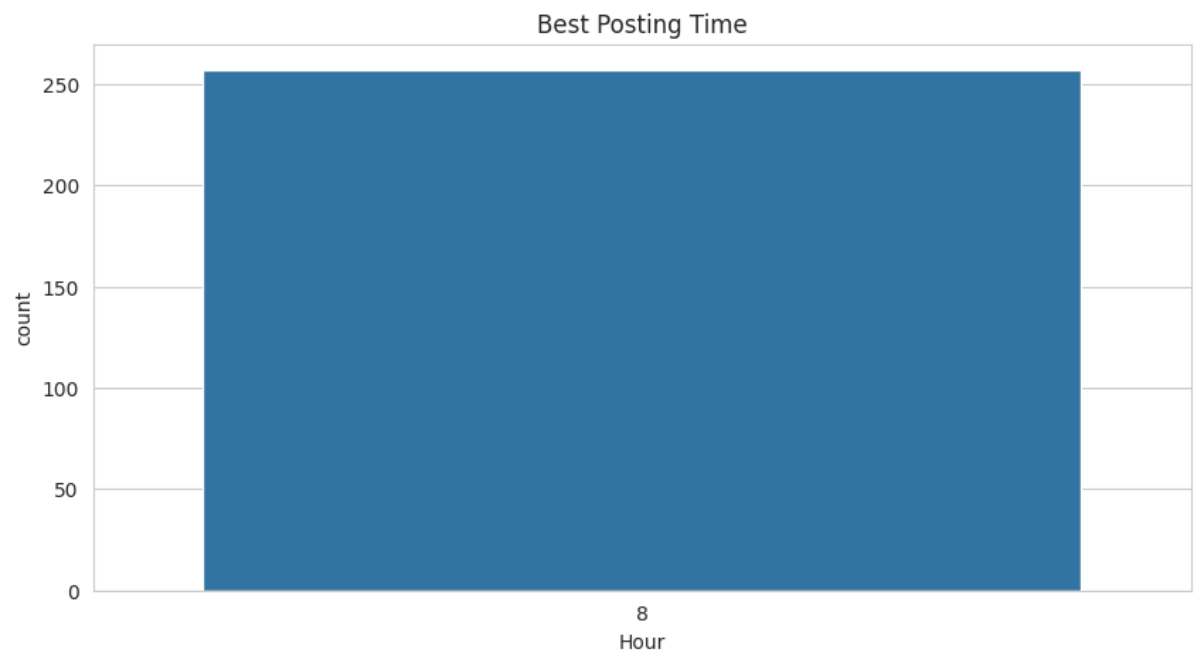


Fig 1.9:Best Posting Time

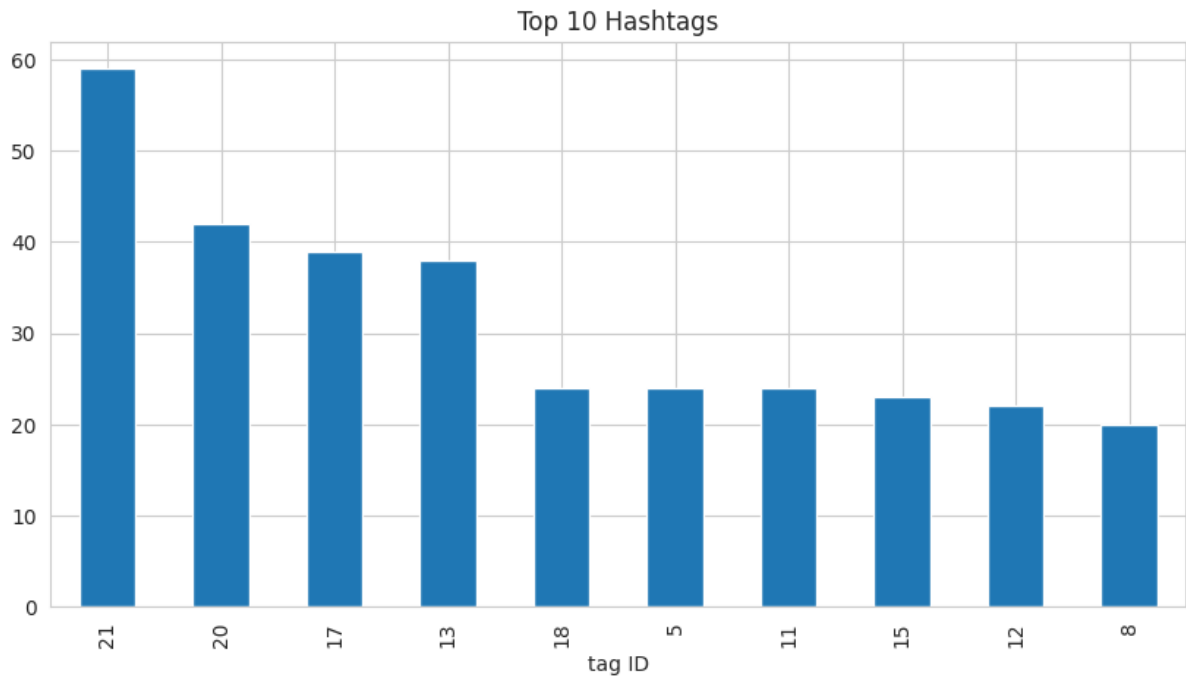


Fig 1.10:Top 10 Hashtags

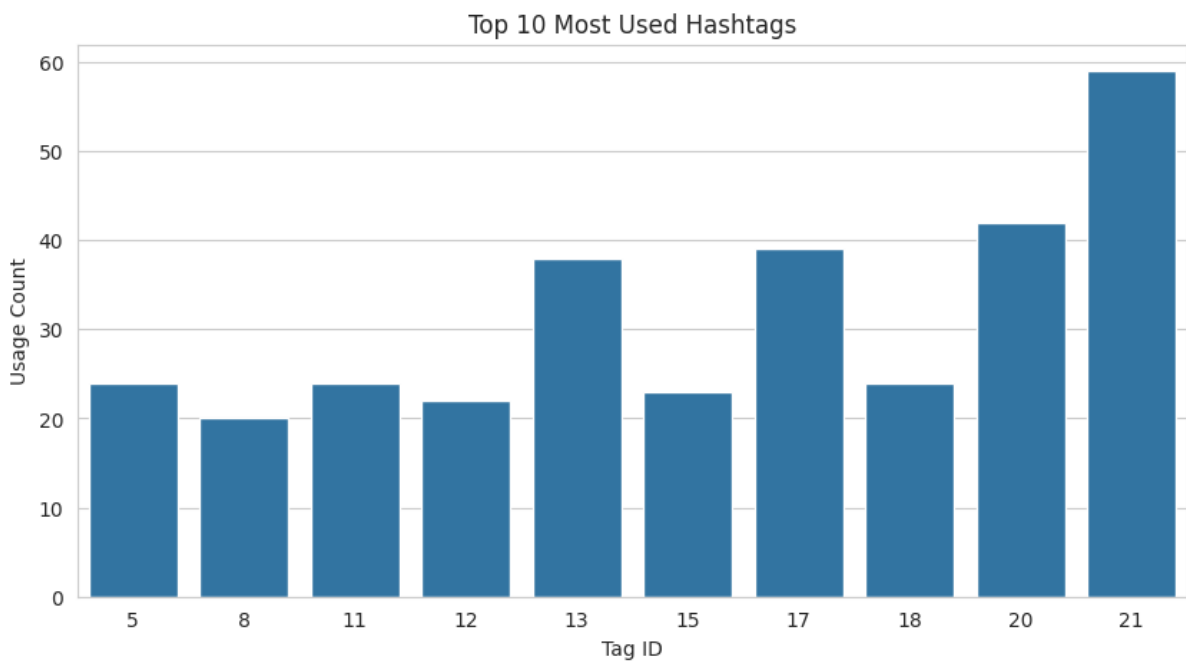


Fig 1.11:Top 10 Most Used Hashtags

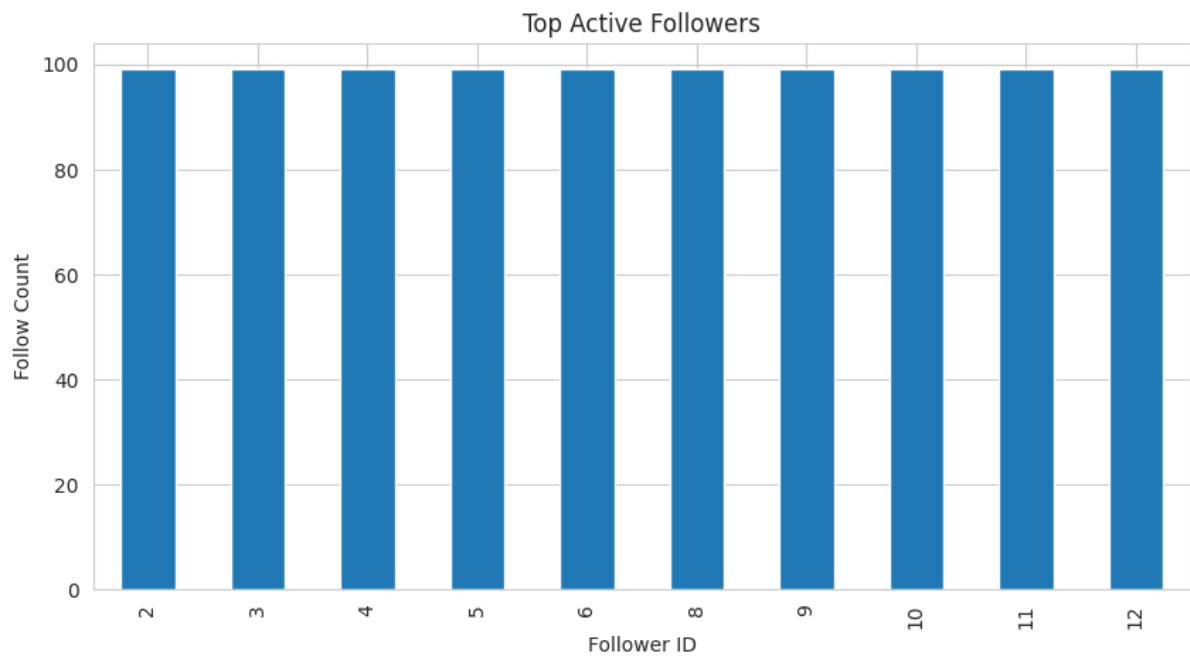


Fig 1.12:Top Active Followers



Fig 1.13:Verified vs Private/Public Accounts

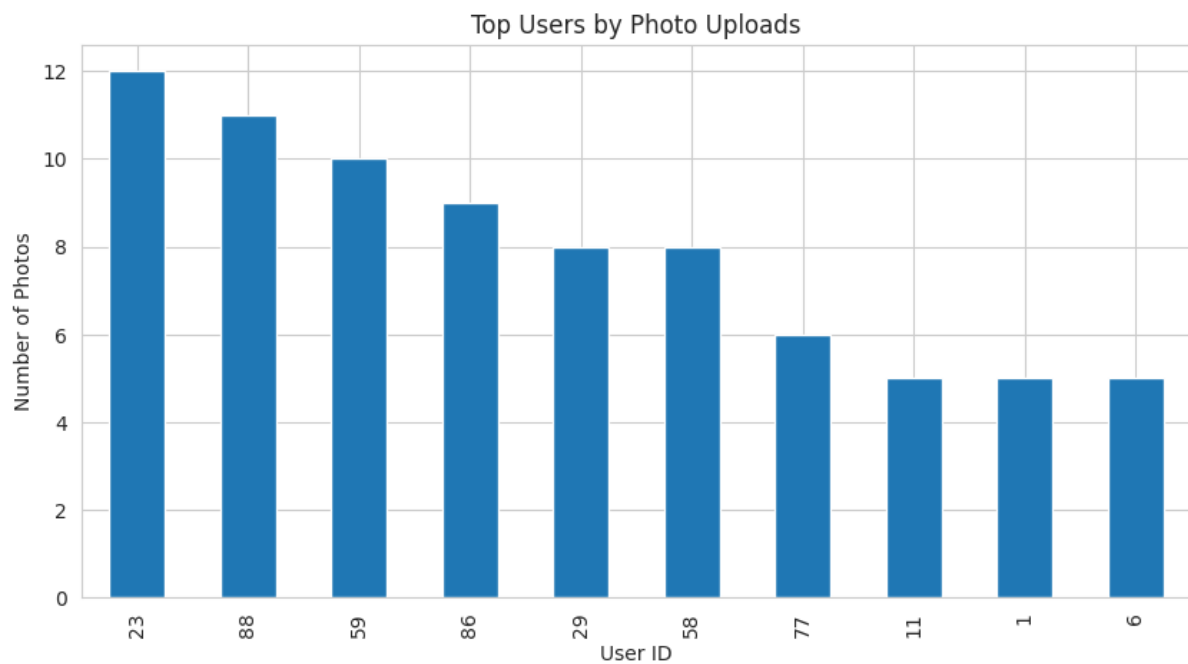


Fig 1.14:Top Users by Photo Uploads

Project Summary

The Instagram Data Analysis project was carried out to understand user behavior and engagement patterns using Instagram datasets. The project involved analyzing seven related datasets containing information about users, photos, likes, comments, followers, hashtags, and tagged photos. Initially, the datasets were imported into Google Colab using the Pandas library. Data preprocessing techniques such as removing duplicate records, checking missing values, and converting date columns into datetime format were applied to improve data quality.

Exploratory Data Analysis (EDA) was performed using Matplotlib and Seaborn to visualize user activity and content performance. Various charts were created to identify the most active users, photo type distribution, hashtag usage, likes, comments, and follower activity. The analysis showed that photo posts were more common than video and carousel posts. A small group of users contributed a significant portion of the total posts, while certain hashtags appeared more frequently than others, indicating trending topics.

The project also highlighted user engagement through likes and comments, providing valuable insights into content popularity. These findings can help organizations understand audience preferences and improve their social media strategies. Based on the analysis, several recommendations were provided for Alfido Tech, including using trending hashtags, posting consistently, creating engaging content, and maintaining an active online presence.

Overall, this project demonstrates how data analytics can be used to improve social media performance and support data-driven decision-making for digital marketing.

Key Insights

- Photo posts are more popular than video and carousel posts.
- A few users contribute a large percentage of total posts.
- Some hashtags are used much more frequently than others.
- User engagement depends mainly on likes, comments, and consistent posting.
- Verified and public accounts generally receive better visibility and engagement.
- Regular posting and effective hashtag usage improve audience interaction.

Business Recommendations for Alfido Tech Platform

1. Maintain a consistent posting schedule to keep the audience engaged.
2. Use trending and relevant hashtags to increase post visibility.
3. Publish more high-quality photo and carousel content to attract users.
4. Encourage user interaction through questions, polls, and engaging captions to increase comments and likes.
5. Monitor engagement metrics regularly and optimize content based on audience preferences and performance.

Conclusion

This project successfully analyzed Instagram user activity using multiple datasets and identified important engagement patterns. The analysis provided insights into user behavior, content performance, hashtag usage, and posting trends. The visualizations and findings can help Alfido Tech improve its social media strategy, increase audience engagement, and make informed marketing decisions. Overall, the project demonstrates the importance of data analysis in enhancing digital marketing performance and business growth.